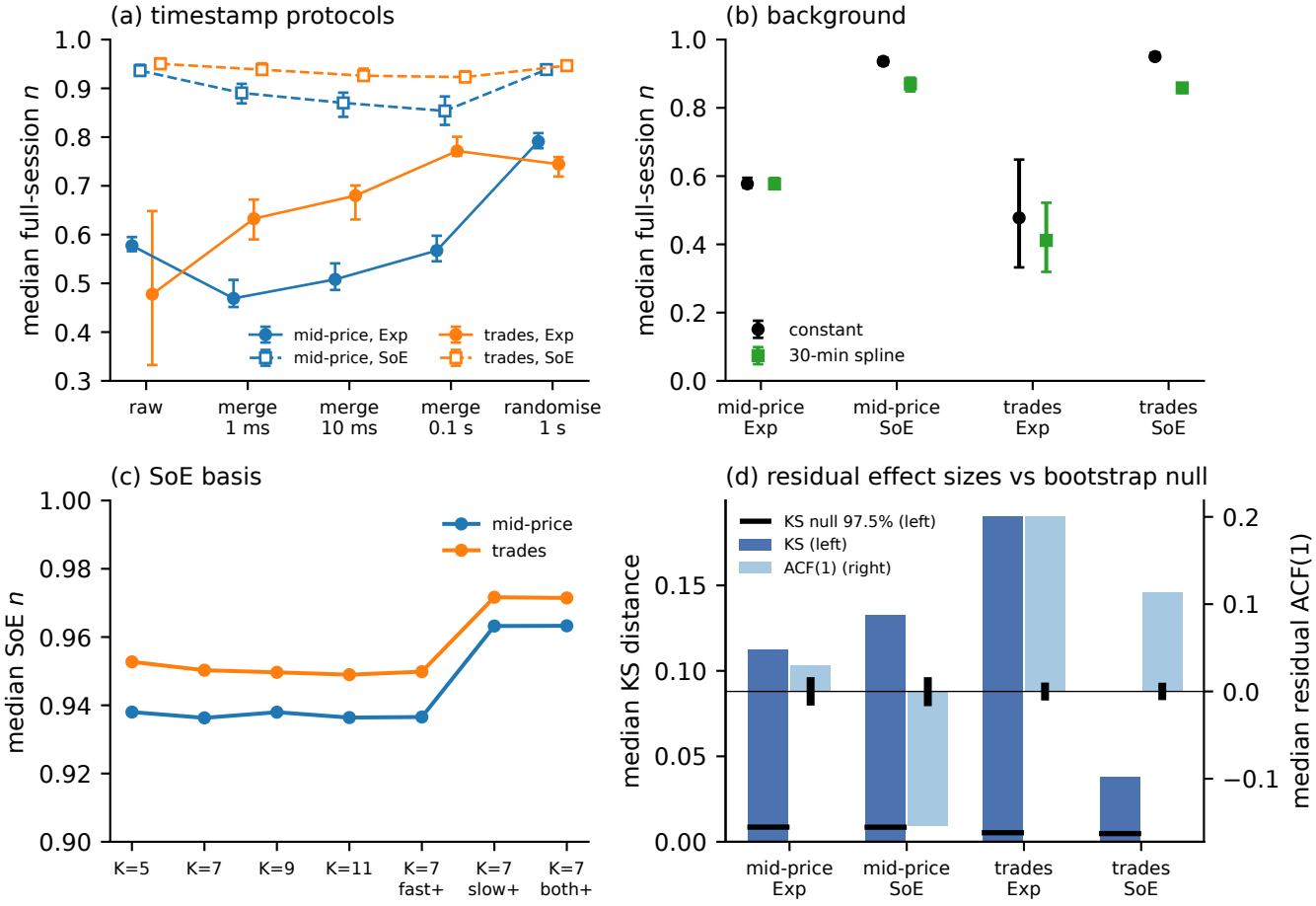
(a) timestamp protocols
median full-session n
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
mid-price, Exp
mid-price, SoE
trades, Exp
trades, SoE
raw
merge 1 ms
merge 10 ms
merge 0.1 s
randomise 1 s
(b) background
median full-session n
1.0
0.8
0.6
0.4
0.2
0.0
constant
30-min spline
mid-price Exp
mid-price SoE
trades Exp
trades SoE
(c) SoE basis
median SoE n
1.00
0.98
0.96
0.94
0.92
0.90
mid-price
trades
K=5
K=7
K=9
K=11
K=7 fast+
K=7 slow+
K=7 both+
(d) residual effect sizes vs bootstrap null
median KS distance
0.15
0.10
0.05
0.00
KS null 97.5% (left)
KS (left)
ACF(1) (right)
0.2
0.1
0.0
−0.1
median residual ACF(1)
mid-price Exp
mid-price SoE
trades Exp
trades SoE